The Growth Rate Hypothesis

Cancer research is one area where stoichiometry is now being employed. Evidence is growing for a theory called the Growth Rate Hypothesis (GRH), which may help explain why some cancerous tumors grow at faster rates than the rest of the body.

The hypothesis states that organisms with high C:P (carbon:phosphorus) ratios, such as fruit flies, have more ribosomes in their cells, which enables them to grow and reproduce more rapidly. Around half of all phosphorus in an organism is in the form of ribosomal RNA (rRNA); it is present in every cell, creating proteins to build new cells and grow the body. Applying biological stoichiometry, James Elser and his team have shown that fast-growing tumors have a much higher phosphorus content than normal body tissue. Such research may help scientists understand how tumor growth could be controlled.